OPENNESS OF THE FLAT LOCUS FOR A FINITE TYPE MODULE OVER A LOCALLY FINITELY PRESENTED MORPHISM WITHOUT NOETHERIAN HYPOTHESES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- A finite type module flat at a point is finitely presented there, and a finitely presented module can be manufactured from the stalk. Same chapter flat.tex, label lemma-finite-type-flat-at-point-local (proof line 4121); Algebra, label algebra-lemma-construct-fp-module-from-stalk (proof line 4123). Produces a finitely presented module F' with a map to F that is an isomorphism at the chosen point, which is the first construction of the proof.
- Openness of flatness and surjectivity of a map of finite type modules detected on a stalk. More on Morphisms, label more-morphisms-theorem-openness-flatness (proof line 4129); Modules, label modules-lemma-finite-type-surjective-on-stalk (proof line 4133); Algebra, label algebra-lemma-NAK (proof line 4135). Makes F' flat over the base after shrinking the source and the map from F' to F surjective after shrinking again, so that the proof is reduced to controlling the kernel.
- Weakly associated points of a tensor product with a pulled back sheaf, finiteness of the associated points of a fibre, and weakly associated primes of a submodule. Same chapter flat.tex, label lemma-bourbaki-finite-type-general-base (proof line 4137); Divisors, labels divisors-lemma-finite-ass (proof line 4145) and divisors-lemma-ses-weakly-ass (proof line 4153); Algebra, labels algebra-lemma-silly (proof line 4147) and Divisors, label divisors-lemma-weakly-ass-zero (proof line 4158). The first bounds the weakly associated points of F' by the finitely many weakly associated points of the base times the associated points of the fibres, the second makes those finite, the third lets the weakly associated points be assumed to specialize

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into the chosen point, and the last two are used to conclude that a module whose weakly associated points are all generalizations of a point at which it vanishes is zero.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Navigation pointer for \ref{algebra-lemma-NAK} used inside the counted span (line 4135); the argument that consumes it is retained in the span.

Source reference. Original lemma 05GJ: sources/stacks-chow/tags.txt, source lines 7063–7063. Navigation pointer for \ref{algebra-lemma-construct-fp-module-from-stalk} used inside the counted span (line 4123); the argument that consumes it is retained in the span.

Source reference. Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Navigation pointer for \ref{algebra-lemma-silly} used inside the counted span (line 4147); the argument that consumes it is retained in the span.

Source reference. Original lemma 05AF: sources/stacks-chow/tags.txt, source lines 6848–6848. Navigation pointer for \ref{divisors-lemma-finite-ass} used inside the counted span (line 4145); the argument that consumes it is retained in the span.

Source reference. Original lemma 05AN: sources/stacks-chow/tags.txt, source lines 6856–6856. Navigation pointer for \ref{divisors-lemma-ses-weakly-ass} used inside the counted span (line 4153); the argument that consumes it is retained in the span.

Source reference. Original lemma 05AP: sources/stacks-chow/tags.txt, source lines 6857–6857. Navigation pointer for \ref{divisors-lemma-weakly-ass-zero} used inside the counted span (line 4158); the argument that consumes it is retained in the span.

Source reference. Original lemma 05IJ: sources/stacks-flat/pinned-original-source.tex, source lines 4079–4079. Navigation pointer for \ref{lemma-bourbaki-finite-type-general-base} used inside the counted span (line 4137); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I7: sources/stacks-flat/pinned-original-source.tex, source lines 2879–2879. Navigation pointer for \ref{lemma-finite-type-flat-at-point-local} used inside the counted span (line 4121); the argument that consumes it is retained in the span.

Source reference. Original lemma 01B8: sources/stacks-chow/tags.txt, source lines 1717–1717. Navigation pointer for \ref{modules-lemma-finite-type-surjective-on-stalk} used inside the counted span (line 4133); the argument that consumes it is retained in the span.

Source reference. Original lemma 0399: sources/stacks-chow/tags.txt, source lines 4264–4264. Navigation pointer for \ref{more-morphisms-theorem-openness-flatness} used inside the counted span (line 4129); the argument that consumes it is retained in the span.

Theorem 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Assume*

(1) *$X \to S$ is locally of finite presentation,*
(2) *$\mathcal{F}$ is an $\mathcal{O}_X$-module of finite type, and*
(3) *the set of weakly associated points of S is locally finite in S.*

Then $U = \{x \in X \mid \mathcal{F}$ flat at x over $S\}$ is open in X and $\mathcal{F}|_U$ is an $\mathcal{O}_U$-module of finite presentation and flat over S.

Proof. Let $x \in X$ be such that $\mathcal{F}$ is flat at x over S. We have to find an open neighbourhood of x such that $\mathcal{F}$ restricts to a S-flat finitely presented module on this neighbourhood. The problem is local on X and S, hence we may assume that X and S are affine. As $\mathcal{F}_x$ is a finitely presented $\mathcal{O}_{X,x}$-module by Lemma 05I7 we conclude from Algebra, Lemma 05GJ there exists a finitely presented $\mathcal{O}_X$-module $\mathcal{F}'$ and a map $\varphi : \mathcal{F}' \to \mathcal{F}$ which induces an isomorphism $\varphi_x : \mathcal{F}'_x \to \mathcal{F}_x$. In particular we see that $\mathcal{F}'$ is flat over S at x, hence by openness of flatness More on Morphisms, Theorem 0399 we see that after shrinking X we may assume that $\mathcal{F}'$ is flat over S. As $\mathcal{F}$ is of finite type after shrinking X we may assume that φ is surjective, see Modules, Lemma 01B8 or alternatively use Nakayama's lemma (Algebra, Lemma 00DV). By Lemma 05IJ we have

$$\mathrm{WeakAss}_X(\mathcal{F}') \subset \bigcup\nolimits_{s \in \mathrm{WeakAss}(S)} \mathrm{Ass}_{X_s}(\mathcal{F}'_s)$$

As $\mathrm{WeakAss}(S)$ is finite by assumption and since $\mathrm{Ass}_{X_s}(\mathcal{F}'_s)$ is finite by Divisors, Lemma 05AF we conclude that $\mathrm{WeakAss}_X(\mathcal{F}')$ is finite. Using Algebra, Lemma 00DS we may, after shrinking X once more, assume that $\mathrm{WeakAss}_X(\mathcal{F}')$ is contained in the generalization of x. Now consider $\mathcal{K} = \mathrm{Ker}(\varphi)$. We have $\mathrm{WeakAss}_X(\mathcal{K}) \subset \mathrm{WeakAss}_X(\mathcal{F}')$ (by Divisors, Lemma 05AN) but on the other hand, φ_x is an isomorphism, also $\varphi_{x'}$ is an isomorphism for all $x' \rightsquigarrow x$. We conclude that $\mathrm{WeakAss}_X(\mathcal{K}) = \emptyset$ whence $\mathcal{K} = 0$ by Divisors, Lemma 05AP. $\square$